4.3 Dividing Polynomials

Name _____ Period ____ Date _____

Algebra II 4.3 · Student sheet

I can…

- use long division to divide polynomials.
- use synthetic division to divide polynomials by x − k.
- use the Remainder Theorem to evaluate polynomial functions.

Vocabulary · fill in the missing word

Polynomial long division — _____ a polynomial by another polynomial, the same way as long division of numbers

Synthetic division — a shortcut for dividing a polynomial by x − k using only the _____

Remainder Theorem — when f(x) is _____ by x − k, the remainder is f(k)

Part 1 · Side by side

785 ÷ 4	$(x^2 + 5x + 7) \div (x + 2)$

f(x) = x² + 5x + 7. f(−2) = _____ What do you notice? _____

Part 2 · Notes: fill in the blanks

Long division steps: _____, multiply, _____, bring down.

For synthetic division by x − k, put _____ in the box; use _____ for missing terms.

Remainder Theorem: the remainder of f(x) ÷ (x − k) is _____.

Part 3 · Examples (work them with the class)

Example 1

Divide $(2x^3 + 3x^2 - x + 5) \div (x^2 - x + 1)$.

Example 2

Divide $(x^3 - 3x^2 - 7x + 6) \div (x - 4)$.

Example 3

Divide $(x^3 - 8) \div (x + 2)$.

Example 4

Use synthetic division to evaluate $f(x) = 3x^4 - 2x^2 + x - 5$ when $x = 2$.

Part 4 · Partner practice

Partner 1

Divide $(x^2 + 7x - 2) \div (x - 1)$.

Partner 2

Use synthetic division to find $f(-3)$ for $f(x) = x^3 + 2x^2 - 5$.